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B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Third Semester

21CSS202T - FUNDAMENTALS OF DATA SCIENCE

(For the candidates admitted from the academic year 2021-2022 to 2024-2025)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer ALL Questions

Marks BL CO PO

1. The most important object defined in NumPy is an N-dimensional array type called as _____

A) ndarray	B) narray	1 1 1 2
C) nd_array	D) darray	

2. _____ is performed by the data scientist after acquiring the data.

A) Data Replication	B) Data Integration	1 2 1 2
C) Data Cleansing	D) Data Analysis	

3. _____ is used to convert numpy array to list

A) array.list()	B) array.list	1 1 1 2
C) list.array()	D) list(array)	

4. Binomial Distribution is a _____.

A) Continuous distribution	B) Discrete distribution	1 2 1 2
C) Irregular distribution	D) Not a Probability distribution	

5. What is the purpose of using 'c' in a pdb command?

A) Display variable	B) Print variable	1 1 2 2
C) Continue	D) Clear the variable	

6. The database management system in SQLite is an ____ database.

A) Embedded relational	B) Non-embedded relational	1 2 2 2
C) Non-relational	D) Structural	

7. What does the function re.search() do?

A) matches a pattern at the start of the string	B) matches a pattern at any position in the string	1 1 2 2
C) such a function does not exist	D) matches a pattern only at the end of the string	

8. JSON Array elements can be accessed by _____.

A) Name	B) Key	1 2 2 2
C) Value	D) Index	

9. Django is written _____ language

A) PHP	B) Python	1 1 3 2
C) Java	D) Perl	

10. What does MVT stand for in Django? 1 2 3 2
 A) model-view-template B) make-test-views
 C) model-template-verify D) mobile-template-verification
11. Which is the correct command to start the Django development server on your system? 1 1 3 2
 A) py manage.py localhost B) py manage.py runatserver
 C) py manage.py createserver D) py manage.py runserver
12. Which is the correct statement to run migrate command in Django? 1 2 3 2
 A) run manage.py migrate B) py migrate
 C) py manage.py migrate D) py manage.py pymigrate
13. What will be the output of following code? 1 1 4 2

```
import pandas as pnd
pnd.Series([1,2], index= ['a','b','c'])
```

 A) Syntax Error B) Segmentation Error
 C) Index Error D) Value Error
14. Which of the following is not a correct way to access individual item from dataframe df? 1 2 4 2
 A) df.iat[2,2] B) df.loc[2,2]
 C) df.at[2,2] D) df[0,0]
15. _____ is used to count non NaN values of dataframe. 1 1 4 2
 A) size B) len
 C) count D) values
16. Which function is used to fill in NaN values with 10 (Assume df is the dataframe object)? 1 2 4 2
 A) fillna(10) B) df.fillna(10)
 C) df.fillna(0) D) df.fill(10)
17. Which of the following is the appropriate syntax for displaying title on a chart? 1 1 5 2
 A) plt.title = "My Title" B) plt.title() = "My Title"
 C) plt.title["My Title"] D) plt.title ("My Title")
18. The default bin value for his() is _____ 1 2 5 2
 A) 5 B) 10
 C) 15 D) 20
19. Which of the following is used to set the limit of X-axis of the plot? 1 1 5 2
 A) Limit() B) Limits()
 C) Xlimit() D) Xlimits()
20. Matplotlib is a _____ library for the Python programming language. 1 2 5 2
 A) Data science B) Mathematics
 C) Numpy D) Plotting

PART - B (5 x 8 = 40 Marks)
 Answer ALL the Questions

Marks BL CO PO

- 21 a. (i) Show the differences between Data Lake and Datawarehouse (4 marks) 8 3 1 2
 (ii) Write notes on facets of data. (4 marks)
- (OR)**
- b. Given a list of tuples, where each tuple represents a student's ID and their corresponding grade, write a NumPy program to sort the students by their grades (descending order) and return their indices. If two students have the same grade, they should be sorted by their ID numbers (ascending order). 8 3 1 2

22 a. Illustrate in detail about unit test module to demonstrate Python debugging. 8 3 2 2

(OR)

b. Write a python program with classes and objects to print name and age of three students by satisfying the following conditions: 8 3 2 2

i) The code must invoke the function `def __init__()`

ii) The printing must be done in the following format with `fun_print()`

My name is:

My age is:

Provide proper comments for every construct and name the variables with readability

23 a. Operate in detail about Django MVT structure with a neat diagram. 8 3 3 2

(OR)

b. Construct the process of URL mapping in DJANGO with necessary code. 8 3 3 2

24 a. Demonstrate different arithmetic, logical, and filtering operations on dataframes. 8 3 4 2

(OR)

b. Interpret the general techniques for handling large volume of data in detail. 8 3 4 2

25 a. Predict the different Univariate visualization for categorical data with assumption of a relevant dataset. 8 3 5 2

(OR)

b. Show the different charts to visualize about how data points values are distributed. 8 3 5 2

PART - C (1 x 15 = 15 Marks)

Marks BL CO PO

Answer ANY ONE Questions

26 i) Prepare a table "DETAILS" and insert the following details in it using SQLite. Also, set ID as the primary key and enforce the condition that fields should not contain NULL values. (5 marks) 15 4 2 2

ID	Name	Gender	Age
1	Sona	Female	20
2	Meena	Female	24
3	Arun	Male	21
4	Mithra	Female	23

ii) Change the name of 'Arun' as 'Varun'. (2 marks)

iii) This table is intended to contain the details of 'Female' only. Change the table accordingly. (2 marks)

iv) Also, insert the following details using different INSERT statements. (3 marks)

ID	Name	Gender	Age
1	Priya	Female	35
2	Kavya	Female	31

v) Illustrate the CRUD operations in SQLite. (3 marks)

27 Demonstrate a program to create a registration form that collects user information (username, email, password, confirm password) and includes validation for each field. The password validation should ensure it meets certain criteria (e.g., minimum length, complexity). Upon successful submission, display a confirmation message. 15 4 3 2
